

SecuringSkies — LLM-Augmented Situational Awareness for ADS-B Threat Detection

Design Document v0.1 (2026-05-31)

1. Problem Statement

The current SecuringSkies ADS-B dashboard presents threat information through: - Numerical scores (spoof_score, ml_score, persistence counter) - Color-coded markers (green/amber/red aircraft) - Alert banners (“CONFIRMED THREAT”) - Expert-mode data tables (GNSS, sensor health)

Gap: A human operator in a high-stress aviation environment cannot efficiently interpret raw ML confidence values, per-feature reconstruction errors, or cross-sensor consistency metrics. They need contextual, role-appropriate explanations of *what* is happening, *why* the system flagged it, and *what action* is recommended.

The existing GhostCommander platform (securingskies-platform) already solves this for drone telemetry — it takes raw MQTT/JSON streams, feeds them through persona-specific LLM prompts, and produces natural-language SITREPs with audited quality metrics. The architecture is transport-agnostic (MQTT + JSON), making it directly applicable to ADS-B anomaly data.

2. Existing Components

From securingskies-platform (LLM layer)

Component	Role	Reusable?
GhostCommander	MQTT ingestion → LLM → SITREP	☐ Core architecture unchanged
Personas (Pilot/Commander/Analyst)	Role-specific prompt engineering	☐ New personas needed for ADS-B context
DSPy Optimizer	Few-shot prompt training from golden logs	☐ Train on ADS-B threat scenarios
Auditor	Recall, Factuality, Hallucination, Safety scoring	☐ Adapt metrics for aviation domain
Constitutional Guardrails	Prevent hallucination, enforce terminology	☐ New rules for aviation terminology

Component	Role	Reusable?
Black Box Recorder	Forensic JSONL logging	☐ As-is

From ADS-B dashboard (detection layer)

Component	Data produced	Format
ML Inference (GRU)	Per-aircraft anomaly score, per-feature error	sensor-core/ml-anomaly
Heuristic engine	spoof_score, spoof_flags	Embedded in map_update
GNSS Monitor	Per-sensor deviation, alerts, jamming/spoofing flags	sensor-core/gnss-health
Accuracy Monitor	Cross-sensor consistency, velocity errors	sensor-core/accuracy
Continual Eval	Model health, threshold drift, FP rate	sensor-core/ml-health
Persistence gauge	Threat confirmation (0-5 windows)	Client-side state

From WebXR (visualization layer)

Component	Role
3D aircraft scene	Spatial awareness — where are threats?
Sensor platforms	Network health at a glance
GNSS sky dome	Satellite integrity visualization
Click-to-inspect	Drill-down on individual aircraft

3. Proposed Architecture

MQTT BROKER (mosquitto)

Topics:

- +/aircraft, sensor-core/ml-anomaly, sensor-core/gnss-health
- sensor-core/accuracy, sensor-core/ml-health

ADS-B SITREP Agent
(new component)

Existing Dashboard
(map_update → browser)

Subscribes to:

- ml-anomaly
 - gnss-health
 - ml-health
 - +/-aircraft (summary)
- WebXR 3D View
(aircraft + sensors)

Triggers SITREP:

- On threat (persist 3)
- On GNSS alert
- On model degradation
- Periodic (every 60s)

Publishes to:

- sensor-core/sitrep

PRESENTATION LAYER

- Option A: Text panel in dashboard (like sensor cards)
- Option B: Voice synthesis (Web Speech API in browser)
- Option C: Dedicated SITREP page with history
- Option D: Overlay in WebXR (floating text panel)
- Option E: All of the above, persona-selectable

4. ADS-B Personas

4.1 ATC Operator (Air Traffic Control)

Context: Monitoring airspace for unauthorized or anomalous traffic. **Style:** Concise, procedural, uses ICAO phraseology. **Triggers on:** New threat, threat escalation, threat cleared. **Example output:** > TRAFFIC ADVISORY: Suspect track 461FA4, FL168, bearing 059 from EFHK, velocity inconsistent with declared flight plan. Confidence HIGH (5/5 windows). Recommend: verify with adjacent FIR.

4.2 Security Analyst

Context: Investigating potential electronic warfare or spoofing campaigns. **Style:** Technical, references specific features and scores, pattern-oriented. **Triggers on:** Any anomaly, GNSS degradation, model drift. **Example output:** > ANALYSIS: GRU reconstruction error 0.14 on hex 461FA4 (FIN1DF), dominant feature: velocity_drift (62%). Pattern consistent with gradual position manipulation — reported GS 406kt but position-derived velocity shows 380kt divergence over 30s window. Cross-sensor RSSI consis-

tent (seen by 3/3 sensors at expected power). Assessment: kinematic spoofing, not RF replay.

4.3 Duty Commander

Context: Decision authority for escalation, sector closure, military coordination. **Style:** Strategic, focuses on patterns across multiple aircraft, recommends actions. **Triggers on:** Confirmed threats, multi-aircraft anomalies, GNSS-wide events. **Example output:** > SITUATION: 2 confirmed anomalous tracks in sector 090-180, FL300-350, both showing velocity drift pattern. GNSS integrity nominal across all ground sensors. Assessment: localized ADS-B spoofing, not wide-area GNSS jamming. Recommend: notify EFHK TWR, cross-reference with primary radar, maintain current operations.

4.4 System Engineer

Context: Monitoring platform health, model performance, sensor status. **Style:** Metrics-focused, infrastructure-aware. **Triggers on:** Model drift, sensor offline, GNSS degradation, high FP rate. **Example output:** > HEALTH: ML model threshold drifted 2.6x from baseline (0.136 → 0.357). FP rate 0%. Cause: score distribution shift — p95 dropped from 0.03 to 0.01, suggesting traffic pattern change (night hours, fewer aircraft). Action: monitor, no recalibration needed unless FP emerges.

5. Confidence Communication Design

The core research question:

How does a human operator interact with AI confidence in high-stress aviation?

Design principles:

1. Never show raw numbers alone. Confidence is communicated through multiple channels simultaneously: - **Spatial** (WebXR): threat position, proximity to airports, clustering - **Temporal** (persistence gauge): how long has the system been confident? - **Linguistic** (SITREP): natural-language explanation calibrated to role - **Visual** (color/animation): immediate pre-attentive processing

2. Explain the “why” not just the “what”. The LLM bridges ML features → human understanding: - “velocity_drift = 0.62” → “aircraft is reporting speed inconsistent with its position changes” - “spoof_score = 0.8, flags=[rssi_error]” → “signal strength doesn’t match the claimed distance”

3. Calibrated language for confidence levels. | Persistence | ML Score | Language
| |-----|-----|-----| | 1/5 | > threshold | “Anomaly detected, monitoring” | | 3/5 | sustained | “Suspected threat, elevated confidence” | | 5/5 | confirmed | “Confirmed threat, high confidence” | | 5/5 + multi-aircraft | pattern | “Coordinated attack pattern detected” |

4. Distinguish sensor issues from actual threats. The LLM must cross-reference: - Is GNSS degraded? → “Reduced detection confidence due to sensor-east multipath” - Is model drifting? → “Note: ML threshold elevated, detection sensitivity reduced” - Single sensor vs multi-sensor? → “Corroborated by 3/3 sensors” vs “Single-sensor detection only”

5. Action-oriented, not information-oriented. Each SITREP ends with a recommended action appropriate to the persona’s authority level.

6. Integration with WebXR

Concept: “AI Wingman” overlay in 3D view

In the WebXR scene, a floating panel (like the existing HUD) shows the latest SITREP. When a threat is active: - The flagged aircraft gets a pulsing ring (already exists) - A “thought bubble” line connects the aircraft to a floating text card explaining why - The card content is the LLM-generated explanation, not raw numbers - Voice readout via Web Speech API (optional, toggle button)

Data flow:

```
sensor-core/sitrep (MQTT) → browser subscribes → update HUD panel
                                     → trigger TTS if enabled
                                     → highlight referenced aircraft
```

7. Auditor Adaptation for Aviation

The existing auditor measures Recall, Factuality, Hallucination, Safety. For ADS-B:

Metric	Aviation adaptation
Recall	Did the SITREP mention all flagged aircraft?
Factuality	Do stated scores/altitudes/callsigns match the data?
Hallucination	Did it invent aircraft, threats, or sensor states?
Safety	Does it recommend actions within authority? No “shoot down” language?
Timeliness (new)	Was the SITREP generated within 2s of trigger?
Consistency (new)	Does repeated analysis of same data produce same assessment?

8. Implementation Phases

Phase 1: Backend SITREP agent (minimal)

- New service: `adsb_sitrep_agent.py`
- Subscribes to `ml-anomaly`, `gnss-health`, `ml-health`
- On threat trigger: formats context, calls LLM, publishes to `sensor-core/sitrep`
- Single persona (Security Analyst)
- Model: Ollama llama3.1 (local) or litellm for cloud fallback

Phase 2: Dashboard integration

- Add SITREP panel to main dashboard (collapsible, like sensor cards)
- Subscribe to `sensor-core/sitrep` via Socket.IO
- Show latest SITREP with timestamp and confidence level

Phase 3: Multi-persona + WebXR

- Persona selector in UI
- WebXR floating SITREP panel
- Voice synthesis toggle
- DSPy optimization using recorded threat events as training data

Phase 4: Research evaluation

- A/B test: operators with/without LLM SITREP
 - Measure: time-to-decision, false alarm response rate, situational awareness score
 - Publish findings on human-AI teaming in aviation security
-

9. Key Design Decisions (TBD)

1. **Local vs cloud LLM?** Latency requirement is <2s for threat response. Llama 3.1 8B on local GPU vs API call?
 2. **Trigger frequency?** Every anomaly? Only on persistence ≥ 3 ? Periodic summary?
 3. **How much context to feed?** Full aircraft list + all scores? Or just the flagged aircraft?
 4. **Voice in WebXR?** Spatial audio (threat direction) or standard TTS?
 5. **Multi-language?** Finnish/English for ATC context?
 6. **Liability framing?** “AI recommends” vs “AI observes” — EU AI Act implications for aviation safety systems.
-

10. Relationship to Existing Work

This design bridges two existing systems without modifying either: - **securingskies-platform**: Proven LLM-MQTT-SITREP architecture (drone domain) - **adsb-dashboard**: Proven ML anomaly detection (ADS-B domain)

The new component (`adsb_sitrep_agent`) is a subscriber/publisher that reads from one and writes for the other. Both existing systems continue to operate independently.

The research contribution is the **confidence communication framework** — how to translate ML model outputs into actionable human understanding in safety-critical aviation contexts, using role-specific LLM personas with audited quality guarantees.